

Notes: Goldberg & Maggi (AER, 1999)

Protection for Sale: An Empirical Investigation

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	Protection measure	3
3.2	Political organization	3
3.3	Import penetration and elasticity	3
3.4	Additional controls	3
4	Theory and econometric specification	4
4.1	Government and lobby objectives	4
4.2	Protection equation from the Grossman–Helpman model	4
4.3	Structural interpretation	4
5	Identification and empirical design	4
5.1	What identifies the model	4
5.2	Why the test is not a standard causal design	5
5.3	Main threats	5
5.4	Responses and robustness	5
6	Tables and figures	6
6.1	Table 1: Results from the Basic Specification; Table 2: Alternative Specifications	6
6.2	Table 3: Results from Using Alternative Thresholds; Table 4: Results from Interacting Contribution Levels with Import Penetration	6
6.3	Table A1: Reduced-Form Equation for Import Penetration; Table A2: Reduced-Form Equation for Political Organization	7
6.4	Table A3: Results from an Extended Specification; Table B1 and Figure B1: Definition of Organization Dummy	8
7	Short summary	8
8	Conclusion	9
8.1	Core conclusion	9
8.2	Economic intuition	9
8.3	Incremental contribution revisited	9
8.4	Interpretation of the high welfare weight	9
8.5	Limitations	9
8.6	Takeaway	10

1 Big picture

- **Question.** Do the cross-sectional patterns of U.S. import protection fit the Grossman–Helpman “Protection for Sale” model, and what do the data imply about the government’s weight on welfare relative to political contributions?
- **Theory being tested.** In the Grossman–Helpman model, organized industries lobby for protection. Equilibrium protection depends on import-demand elasticity, import penetration, and whether an industry is politically organized.
- **Core empirical prediction.** Protection should be lower when import demand is more elastic. The effect of import penetration depends on organization: protection rises with X_i/M_i for organized industries and falls with X_i/M_i for unorganized industries.
- **Data.** The paper uses U.S. manufacturing industries in 1983, nontariff-barrier coverage ratios as the protection measure, import-demand elasticities, output/import ratios, and PAC contributions to classify political organization.
- **Main result.** The signs of the estimated coefficients match the Grossman–Helpman predictions. The basic model is not improved much by standard reduced-form protection variables.
- **Structural result.** The implied government welfare weight is very high, around 0.98, while the contribution weight is small. Still, the pure welfare-maximization/free-trade restriction is rejected.
- **Interpretation.** Politics matters for the cross-sectional pattern of trade barriers, even though welfare receives far more weight than contributions in the estimated government objective.
- **Caveat.** The protection measure is an NTB coverage ratio rather than an ad valorem tariff equivalent. This is a serious measurement limitation, especially for the structural magnitudes.

2 Incremental contribution of the paper

1. **First structural empirical test of a leading political-economy trade model.** Earlier empirical work on protection was largely reduced form. This paper lets the theoretical Grossman–Helpman model dictate the regressors, signs, and functional form.
2. **Structural parameter recovery.** The paper estimates not only whether lobbying matters, but also the model-implied welfare weight β and represented-population share α_L .
3. **Interaction logic.** The paper emphasizes a distinctive prediction: import penetration has different signs for organized and unorganized industries. This goes beyond simply asking whether politically powerful industries receive more protection.
4. **Political organization measurement.** The paper uses PAC contributions to construct an organized-industry dummy, connecting theory’s lobbying status to observed political money.
5. **Model discipline against ad hoc controls.** The paper compares the strict Grossman–Helpman specification with standard variables from the protection literature and finds little additional explanatory power.
6. **Validation with default rates? No - validation with robustness and alternative thresholds.** Since there is no direct observable counterfactual for protection, the paper validates the model through alternative organization definitions, contribution-level interactions, exogeneity checks, and different treatment of the dependent variable.

Interpretation: The contribution is to move the empirical political-economy literature from “which variables correlate with protection?” to “does a particular lobbying model quantitatively match the protection schedule?”

3 Data and measurement

3.1 Protection measure

The dependent variable is based on nontariff-barrier coverage ratios for U.S. manufacturing sectors in 1983. A coverage ratio is the import-share-weighted fraction of products in an industry covered by some nontariff barrier.

- The theory is written for tariffs; the data use NTB coverage ratios.
- Coverage ratios are treated as proxies for the ranking of tariff equivalents across sectors.
- The average coverage ratio is low relative to its maximum, but it varies substantially across sectors.

Interpretation: The paper’s qualitative sign tests are more robust than the quantitative structural estimates. The structural estimates require stronger assumptions about the relationship between coverage ratios and true protection levels.

3.2 Political organization

An industry is classified as politically organized using PAC contributions by 3-digit SIC industry. In the baseline, sectors above the chosen contribution threshold are organized; sectors below it are unorganized. Appendix B lists unorganized sectors and displays the contribution distribution.

Interpretation: The organization dummy is the empirical counterpart of lobbying status in the Grossman–Helpman model. The model’s interaction term $X_i/M_i \times I_i$ is meaningful only because I_i has a concrete contribution-based definition.

3.3 Import penetration and elasticity

The key economic variables are:

- X_i/M_i : output-to-import ratio, inverse of import penetration in the paper’s terminology;
- e_i : import-demand elasticity;
- $t_i/(1 + t_i)$: transformed protection measure.

Interpretation: The model predicts that protection is more costly in elastic sectors and that organized sectors receive more protection when the domestic-output-to-import ratio is high.

3.4 Additional controls

The paper tests whether standard variables from earlier protection regressions add explanatory power: unemployment, employment size, unionization, concentration measures, changes in import penetration, tenure, and industry growth.

Interpretation: These controls are not in the strict Grossman–Helpman model. Their role is diagnostic: if they matter strongly, the Grossman–Helpman model is incomplete.

4 Theory and econometric specification

4.1 Government and lobby objectives

Aggregate welfare is denoted W . Organized lobbies care about their industry returns plus consumer/tax-transfer effects. The government objective combines social welfare and political contributions:

$$U^G = \beta W + (1 - \beta) \sum_{i \in L} C_i. \quad (1)$$

Here $\beta \in [0, 1]$ is the weight on welfare, and $1 - \beta$ is the weight on contributions.

Interpretation: A high β does not imply lobbying is irrelevant. Even a small contribution weight can shape the cross-industry allocation of protection when lobbies target sectors selectively.

4.2 Protection equation from the Grossman–Helpman model

The theory implies:

$$\frac{t_i}{1 + t_i} = \frac{I_i - \alpha_L}{\beta/(1 - \beta) + \alpha_L} \frac{z_i}{e_i}, \quad z_i \equiv X_i/M_i. \quad (2)$$

Multiplying by e_i and adding an error gives the estimating equation:

$$\frac{t_i}{1 + t_i} e_i = \gamma \frac{X_i}{M_i} + \delta I_i \frac{X_i}{M_i} + \varepsilon_i. \quad (3)$$

The structural parameters imply:

$$\gamma < 0, \quad \delta > 0, \quad \gamma + \delta > 0. \quad (4)$$

Interpretation: Unorganized sectors have slope $\gamma < 0$. Organized sectors have slope $\gamma + \delta > 0$. This is the key sign pattern.

4.3 Structural interpretation

The reduced-form coefficients map into the two structural parameters:

- β : the welfare weight in the government’s objective;
- α_L : the share of the population represented by lobbies.

Interpretation: The same regression estimates that test the sign predictions also identify the model-implied welfare weight. This is why the paper is more structural than earlier protection regressions.

5 Identification and empirical design

5.1 What identifies the model

The model is identified from cross-industry variation in:

- import protection;
- import-demand elasticity;
- output-to-import ratio;

- political organization status.

The decisive empirical object is the interaction $I_i \times X_i/M_i$. The Grossman–Helpman model predicts different protection-penetration slopes depending on whether the sector is organized.

5.2 Why the test is not a standard causal design

This is not a natural experiment in lobbying. The paper does not randomly assign political organization. Instead, it asks whether the equilibrium cross-sectional pattern of protection satisfies the structural restrictions of the Grossman–Helpman model.

Interpretation: The question is model validation and structural estimation, not a causal treatment effect of lobbying.

5.3 Main threats

- **Protection measurement.** NTB coverage ratios are imperfect proxies for tariff equivalents.
- **Endogeneity.** Import penetration and political organization may be endogenous to the protection process.
- **Measurement error.** Import-demand elasticities are noisy.
- **Organization threshold.** The PAC-contribution cutoff used to define organized sectors is partly arbitrary.
- **Model scope.** The model is static and does not explicitly model unemployment, unionization, or sectoral political institutions.

5.4 Responses and robustness

The paper addresses these concerns by estimating reduced-form equations for import penetration and political organization, using alternative organization thresholds, interacting contribution levels directly with import penetration, testing standard control variables, and checking alternative definitions of the dependent variable.

Interpretation: The strongest case for the model is that the core signs and structural implications survive these robustness checks, while additional reduced-form variables add little explanatory power.

6 Tables and figures

6.1 Table 1: Results from the Basic Specification; Table 2: Alternative Specifications

TABLE 1—RESULTS FROM THE BASIC SPECIFICATION
(G-H MODEL)

Variable	$\mu = 1$	$\mu = 2$	$\mu = 3$
X_i/M_i	−0.0093 (0.0040)	−0.0133 (0.0059)	−0.0155 (0.0070)
$(X_i/M_i) * I_i$	0.0106 (0.0053)	0.0155 (0.0077)	0.0186 (0.0093)
Implied β	0.986 (0.005)	0.984 (0.007)	0.981 (0.009)
Implied α_L	0.883 (0.223)	0.858 (0.217)	0.840 (0.214)

TABLE 2—ALTERNATIVE SPECIFICATIONS ($\mu = 1$)

Variable	Specification 1 Log-likelihood: −134.9	Specification 2 Log-likelihood: −132.06	Specification 3 Log-likelihood: −132.04	Specification 4 Log-likelihood: −130.61
X_i/M_i	—	−0.0093 (0.0040)	−0.0096 (0.0043)	−0.0109 (0.0045)
$(X_i/M_i) * I_i$	—	0.0106 (0.0053)	0.0105 (0.0053)	0.0123 (0.0055)
Constant	−0.0640 (0.1104)	—	−0.0287 (0.1375)	−0.2619 (0.2559)
Unemployment	—	—	—	1.5722 (1.5884)
Employment size	—	—	—	1.1836 (0.8235)

Note: Dependent variable: $(r_i^p e_i / 1 + r_i^p)$.

(a) Table 1: basic Grossman–Helpman specification.

(b) Table 2: alternative specifications.

Table 1 and Table 2. Table 1 reports the core Grossman–Helpman regression for alternative values of μ ; Table 2 adds standard reduced-form controls. The signs match the model: X_i/M_i is negative for unorganized sectors and the interaction with organization is positive. The additional controls barely improve fit.

Read:

- In Table 1, X_i/M_i is negative and $(X_i/M_i) \times I_i$ is positive.
- The implied welfare weight β is around 0.98, while α_L is around 0.84–0.88.
- Table 2 shows that unemployment and employment size have intuitive signs but do not overturn the strict model.

Economic message: The basic model fits the sign pattern, and the strict theory is not dominated by the usual ad hoc controls. The high welfare weight explains why average protection remains low. The model is still useful because the small contribution weight shapes cross-sectoral differences.

6.2 Table 3: Results from Using Alternative Thresholds; Table 4: Results from Interacting Contribution Levels with Import Penetration

TABLE 3—RESULTS FROM USING ALTERNATIVE THRESHOLDS TO DEFINE THE POLITICAL-ORGANIZATION DUMMY

Variable	Thresholds		
	\$50,000,000 Percent of organized sectors: 74	0.1 percent of total contributions Percent of organized sectors: 84	0.1 percent of value added Percent of organized sectors: 85
X_i/M_i	−0.0090 (0.0039)	−0.1475 (0.0664)	−0.0045 (0.0025)
$(X_i/M_i) * I_i$	0.0099 (0.0054)	0.1286 (0.0697)	0.0075 (0.0074)

TABLE 4—RESULTS FROM INTERACTING CONTRIBUTION LEVELS WITH IMPORT PENETRATION

Variable	Coefficient (Standard error)
X_i/M_i	−0.0096 (0.0036)
$(X_i/M_i) * C_i$	0.0032 (0.0013)

(a) Table 3: alternative political-organization thresholds.

(b) Table 4: contribution levels interacted with import penetration.

Table 3 and Table 4. These are robustness checks for the definition of political organization. Table 3 changes the cutoff defining organized sectors; Table 4 replaces the organization dummy with contribution levels. The main signs are stable.

Read:

- Table 3 shows that the signs survive alternative thresholds, though precision falls as the organized share becomes very high.

- Table 4 shows that using continuous contribution levels yields the same qualitative sign pattern.

Economic message: The paper’s result is not purely an artifact of one binary organization cutoff. The continuous-contribution interaction is not the strict Grossman–Helpman model, but it validates the intuition that politically stronger sectors get different protection slopes.

6.3 Table A1: Reduced-Form Equation for Import Penetration; Table A2: Reduced-Form Equation for Political Organization

TABLE A1—REDUCED-FORM EQUATION
FOR IMPORT PENETRATION

Variable	Coefficient	Standard error	<i>t</i>
Physical capital	1296.03	435.28	2.98
Inventories	−2501.81	709.42	−3.53
Engineers and scientists	−52.79	405.38	−0.13
White-collar	489.58	192.54	2.54
Skilled	91.86	252.80	0.36
Semiskilled	−99.89	202.18	−0.49
Cropland	514.86	231.45	2.23
Pasture	−1493.13	366.34	−4.08
Forest	4262.02	3451.79	1.24
Coal	3043.31	3356.99	0.91
Petroleum	5.62	154.04	0.04
Minerals	−4778.79	4020.54	−1.19
Seller concentration	−37.46	56.29	−0.67
Seller number of firms	−5.64	28.03	−0.20
Buyer concentration	−78.17	114.22	−0.68
Buyer number of firms	−2.35	12.76	−0.18
Scale	−10.80	249.05	−0.04
Capital stock	−118.88	42.99	−2.77
Unionization	79.64	46.55	1.71
Geographic concentration	−43.02	44.57	−0.97
Tenure	−0.21	4.73	−0.05
Constant	−11.62	79.70	−0.15

Notes: Dependent variable: Inverse penetration ratio (X_i/M_i).

Number of observations: 107.

R^2 : 0.49.

(a) Table A1: reduced-form import penetration equation.

TABLE A2—REDUCED-FORM EQUATION
FOR POLITICAL ORGANIZATION

Variable	Coefficient	Standard error	<i>t</i>
Physical capital	23.90	15.25	1.57
Inventories	−5.14	33.83	−0.15
Engineers and scientists	21.03	15.73	1.34
White-collar	−7.30	6.71	−1.09
Skilled	22.34	10.97	2.04
Semiskilled	−18.53	7.67	−2.42
Cropland	15.57	21.13	0.74
Pasture	−25.88	35.86	−0.72
Forest	−11.87	123.90	0.10
Coal	136.72	104.36	1.31
Petroleum	10.45	6.06	1.73
Minerals	63.35	214.73	0.30
Seller concentration	−3.32	1.76	−1.88
Seller number of firms	−1.69	1.23	−1.38
Buyer concentration	2.13	4.27	0.50
Buyer number of firms	−0.53	0.56	−0.96
Scale	50.81	18.84	2.70
Capital stock	−2.83	1.51	−1.88
Unionization	2.16	1.58	1.37
Geographic concentration	−3.22	1.62	−1.99
Tenure	−0.44	0.16	−2.73
Constant	2.59	2.92	0.89

Notes: Dependent variable: Political-organization dummy (I_i).

Number of observations: 107

(b) Table A2: reduced-form political organization equation.

Table A1 and Table A2. Appendix A reports the auxiliary equations for import penetration and political organization. These equations address endogeneity and provide inputs for the simultaneous estimation approach.

Read:

- Table A1 shows determinants of inverse import penetration such as physical capital, inventories, and land shares.
- Table A2 shows determinants of political organization, including scale and worker-skill variables.

Economic message: These auxiliary equations are not the main theory test; they help implement the econometric system and deal with endogenous variables.

6.4 Table A3: Results from an Extended Specification; Table B1 and Figure B1: Definition of Organization Dummy

Variable	Coefficient	Standard error	<i>t</i>
X_i/M_i	-0.0092	0.0044	-2.104
$(X_i/M_i) * I_i$	0.0089	0.0089	1.998
Constant	-0.2545	0.2409	-1.057
I_i	0.3851	0.3466	1.111

Note: Dependent variable: $(t_i^*e_i/1 + t_i^*)$.

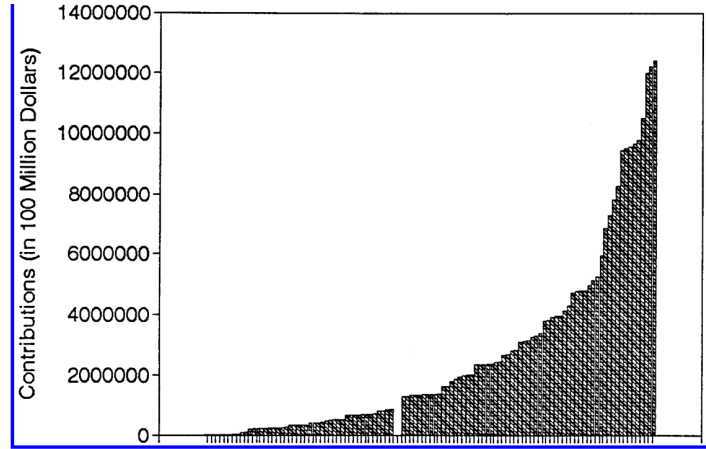


Figure B1: Bar Chart of PAC Contributions by SIC

(a) Table A3: extended specification.

(b) Figure B1: PAC contributions by SIC3.

Table A3 and Figure B1. Table A3 adds the organization dummy directly to the protection equation. Figure B1 visualizes the contribution distribution that motivates the organized/unorganized classification.

TABLE B1—LIST OF UNORGANIZED SECTORS

Code	Standard Industrial Classification (SIC)	Code	Standard Industrial Classification (SIC)
SIC 315	Leather Gloves and Mittens	SIC 227	Floor Covering Mills
SIC 319	Leather Goods, Not Elsewhere Classified	SIC 228	Yarn and Thread Mills
SIC 316	Luggage	SIC 321	Flat Glass
SIC 313	Boot and Shoe Cut Stock and Findings	SIC 328	Cut Stone and Stone Products
SIC 311	Leather Tanning and Finishing	SIC 241	Logging Camps and Logging Contractors
SIC 213	Tobacco (Chewing and Smoking) and Snuff	SIC 251	Household Furniture
SIC 212	Cigars	SIC 324	Cement, Hydraulic
SIC 314	Footwear, Except Rubber	SIC 322	Glass and Glassware, Pressed or Blown
SIC 317	Handbags and Other Personal Leather Goods	SIC 306	Fabricated Rubber Products, Not Elsewhere Classified
SIC 393	Musical Instruments	SIC 301	Tires and Inner Tubes
SIC 231	Men's, Youths', and Boys' Suits, Coats, and Overcoats	SIC 302	Rubber and Plastics Footwear
SIC 259	Miscellaneous Furniture and Fixtures	SIC 385	Ophthalmic Goods
SIC 278	Blankbooks, Looseleaf Binders, and Bookbinding and Related Work	SIC 383	Optical Instruments and Lenses
SIC 273	Books	SIC 387	Watches, Clocks, Clockwork Operated Devices, and Parts
SIC 271	Newspapers: Publishing, Publishing and Printing	SIC 381	Engineering, Laboratory, Scientific, and Research Instruments, and Associated Equipment
SIC 272	Periodicals: Publishing, Publishing and Printing	SIC 386	Photographic Equipment and Supplies
SIC 399	Miscellaneous Manufacturing Industries	SIC 339	Miscellaneous Primary Metal Products
SIC 395	Pens, Pencils, and Other Office and Artists' Materials	SIC 332	Iron and Steel Foundries
SIC 394	Toys and Amusement, Sporting, and Athletic Goods	SIC 249	Miscellaneous Wood Products
SIC 391	Jewelry, Silverware, and Plated Ware	SIC 244	Wood Containers
SIC 275	Commercial Printing	SIC 307	Miscellaneous Plastics Products
SIC 396	Costume Jewelry, Costume Novelties, Buttons, and Miscellaneous Notions, Except Precious Metal	SIC 295	Paving and Roofing Materials

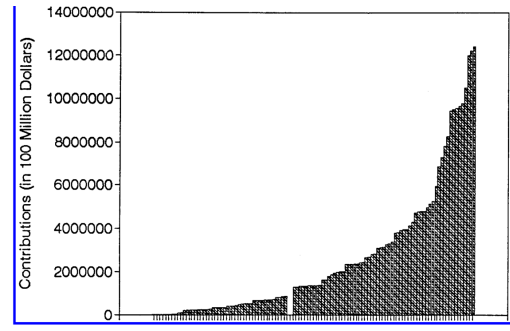


Figure B1: Bar Chart of PAC Contributions by SIC

(a) Table B1: list of unorganized sectors.

(b) Figure B1 repeated for direct comparison.

Table B1 and Figure B1. Table B1 lists sectors treated as unorganized under the baseline rule; Figure B1 shows the skewed PAC-contribution distribution behind the classification.

Read:

- Figure B1 shows a very skewed contribution distribution with a natural separation between low-contribution and high-contribution sectors.
- Table B1 lists sectors with low contribution levels that are classified as unorganized.

Economic message: Political organization is not directly observed, so the contribution distribution is crucial for operationalizing the theory.

7 Short summary

- The paper structurally tests the Grossman–Helpman Protection for Sale model using U.S. manufacturing NTBs in 1983.

- The dependent variable is based on NTB coverage ratios, and the main regressors are import elasticity, output/import ratio, and political organization.
- The key prediction is that the protection-import-penetration relationship has opposite signs for organized and unorganized sectors.
- The signs in the data match the model.
- Additional reduced-form variables add little explanatory power.
- Alternative definitions of political organization and contribution-level interactions preserve the core result.
- The implied government welfare weight is very high, around 0.98, while the contribution weight is small.
- Nevertheless, the pure welfare-maximization/free-trade model is rejected.
- The main limitation is that NTB coverage ratios are imperfect measures of true trade-policy restrictiveness.

8 Conclusion

8.1 Core conclusion

Goldberg and Maggi find that the cross-sectional structure of U.S. nontariff barriers in 1983 is broadly consistent with the Grossman–Helpman Protection for Sale model. Organized industries face a different protection schedule from unorganized industries, and the estimated signs align with the theory.

8.2 Economic intuition

An organized sector’s lobby gains when protection raises the domestic price of its output. The government trades off this political benefit against the welfare cost of protection. Protection is more attractive when import demand is inelastic and when domestic output is large relative to imports. For unorganized sectors, the government’s incentive is driven mainly by the welfare cost, so protection moves differently with import penetration.

8.3 Incremental contribution revisited

The paper’s main contribution is to estimate a political-economy trade model structurally rather than running a reduced-form protection regression. It uses the model’s own predicted interaction between political organization and import penetration, recovers structural welfare weights, and then checks whether standard political-economy variables add explanatory power.

8.4 Interpretation of the high welfare weight

The high estimated welfare weight does not mean lobbying is irrelevant. It means that U.S. trade policy, at least in 1983 manufacturing, is close to welfare maximizing in levels but not in cross-sectional allocation. Political contributions help explain which sectors get more protection, even though protection is generally limited.

8.5 Limitations

The main limitations are measurement and scope. NTB coverage ratios are rough proxies for tariff equivalents; the analysis uses one country and one year; political organization is inferred from contributions; and the model abstracts from many institutional features of trade policy. The structural magnitudes should therefore be interpreted cautiously.

8.6 Takeaway

The paper shows that formal political-economy models can be brought to the data. Its lasting lesson is methodological as much as substantive: a lobbying model should not merely predict that powerful industries are protected; it should generate a disciplined equation that can be estimated and tested against observed protection patterns.